

Execution of the Local Occurrence Hypothesis

Paper 73 — First Test of the Preregistered SAT-Specific Protocol

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2026

Abstract

Paper 72 preregistered a SAT-specific follow-up to Gate Challenge Cycle I: the Local Occurrence Hypothesis. The hypothesis stated that decision-local occurrence/degree residuals might add useful SAT branching information beyond the frozen MAAT v1.7 gate. This paper executes that frozen protocol without changing the definition of $L_\star$, the residualization rule, calibration splits, budgets, bootstrap procedure, baselines, or interpretation criteria.

The dataset gatekeeper passed on 55 public SATLIB/DIMACS benchmark instances with SHA256-verified manifests and parseable DIMACS files. The execution used 11 calibration instances and 44 test instances across six benchmark families. Under the preregistered interpretation rule, Gate+L failed to beat the frozen gate with a positive 95% confidence lower bound ($\Delta = -13.2382$, CI $[-37.3121, 6.0309]$), failed to beat score-with-R robustly ($\Delta = 6.3392$, CI $[-8.5437, 21.7627]$), and failed the shuffled- L null test ($\Delta = -6.8162$, CI $[-20.1598, 6.0438]$). It also lost clearly against MOMS ($\Delta = -84.9530$, CI $[-130.0199, -44.5361]$).

Under the preregistered interpretation criteria, the Local Occurrence Hypothesis receives `no_minimum_support`. Paper 72 remains the preregistered protocol document. Paper 73 reports only that the frozen hypothesis did not pass its first execution test.

1 Purpose and Guardrails

Gate Challenge Cycle I produced a negative external SAT/CDCL result for the frozen MAAT v1.7 gate and motivated a diagnostic question: did the gate miss a decision-local occurrence or degree channel? Paper 72 converted that question into a preregistered hypothesis and explicitly froze the next test before any new Gate+L execution. The cycle archive is recorded in the Research Series II release [4].

This paper is an execution report. It does not introduce a new MAAT version, does not alter the Paper-68 gate, and does not modify the Paper-72 local channel. In particular, it does not tune thresholds, change residualization, adjust weights, redesign the solver policy, or reinterpret the success criteria after seeing the result.

The only allowed interpretation classes are those declared in Paper 72:

`no_minimum_support`, `minimum_support_only`, `strong_sat_support`.

2 Frozen Protocol

The Paper-72 hypothesis introduced a SAT-specific local residual coordinate $L_\star$ derived from occurrence and degree channels after calibration-only residualization against the frozen structural coordinates. In the execution code, the frozen channel is

$$L_\star(F) = \frac{1}{2} [z(D_{\text{res}}) - z(O_{\text{res}})], \quad (1)$$

where O_{res} and D_{res} are the residualized occurrence and degree terms, and the z -standardization is performed using calibration-fold statistics only. This definition is reported here for reproducibility; it is not changed in Paper 73.

The frozen Gate+L wrapper is

$$G_{\text{gate+L}}(F) = \sigma(\text{logit}(G_{\text{gate}}(F)) + L_{\star}(F)), \quad (2)$$

with no free multiplier on $L_{\star}$. This deliberately prevents the execution paper from becoming a hyperparameter search.

3 Data, Licensing, and Reproducibility

The external benchmark source is SATLIB, accessed through the public SATLIB benchmark page [1]. The benchmark page provides DIMACS CNF instances and lists the benchmark families used here, but it does not present an explicit open data license for raw redistribution. For that reason, the repository version of this experiment intentionally excludes raw CNF files and downloaded SATLIB archives. SATLIB benchmark instances are not redistributed in this repository or release. The included manifests contain only source URLs, hashes, and metadata required for reproducibility. Users must obtain the original benchmark instances from SATLIB and cite the SATLIB source.

Table 1: Repository data-policy audit for Paper 73.

Artifact	Repository status	License/data note
Raw SATLIB CNFs	Excluded	Must be obtained from SATLIB
SATLIB archives	Excluded	Not redistributed
Manifest CSV	Included	URLs, hashes, metadata only
Solver CSV/JSON	Included	Derived outputs generated by this run
Figures	Included	Generated from derived outputs
Code and README	Included	Project-owned experiment scaffold

The data gatekeeper verifies manifest completeness, file existence, SHA256 hashes, DIMACS parseability, and absence of unexpected CNF files before solver execution. The executed dataset passed all checks:

Table 2: Dataset gatekeeper result.

Check	Result
Manifest	PASS
CNFs present	PASS
SHA256 verification	PASS
DIMACS parseability	PASS
Unexpected files	PASS
Dataset ready	YES

The run used 55 SATLIB/DIMACS instances: 10 AIM, 10 Dubois, 5 pigeon-hole, 10 flat graph-colouring, 10 UF20-91, and 10 UUF50-218 instances. The frozen split contained 11 calibration instances and 44 test instances.

4 Policies and Metrics

The execution compared the following policies, including a VSIDS-style CDCL baseline following the Chaff tradition [2] and the classical Jeroslow–Wang rule [3]:

{VSIDS, MOMS, Jeroslow–Wang, score-with-R, frozen gate v1.7, Gate+L, Gate+shuffled-L}.

The conflict budget was 5000 and the runtime budget was 30 seconds. Confidence intervals were computed by paired bootstrap with 10000 resamples and seed 72072. For a comparator C , the reported paired utility delta is

$$\Delta(C, \text{Gate+L}) = \text{cost}(C) - \text{cost}(\text{Gate+L}).$$

Thus positive Δ means Gate+L required lower compute cost than the comparator; negative Δ means Gate+L was worse.

5 Results

Table 3: Primary paired comparisons on the 44-instance test split. Positive Δ means Gate+L is better.

Comparison	Mean Δ	CI low	CI high	Gate+L wins	Gate+L losses
Gate+L vs frozen gate v1.7	-13.2382	-37.3121	6.0309	20	22
Gate+L vs score-with-R	6.3392	-8.5437	21.7627	22	20
Gate+L vs MOMS	-84.9530	-130.0199	-44.5361	5	37
Gate+L vs Jeroslow–Wang	-106.8685	-400.5297	64.2660	15	27
Gate+L vs VSIDS	-384.7766	-973.2246	6.0257	18	25
Gate+L vs shuffled-L	-6.8162	-20.1598	6.0438	21	23

The minimum-support rule required Gate+L to beat both frozen gate v1.7 and score-with-R with positive 95% confidence lower bounds, and to survive the shuffled- L null. None of these three conditions was met. The strong-SAT support rule additionally required positive confidence lower bounds against MOMS and Jeroslow–Wang; these conditions were also not met.

Table 4: Policy-level compute-cost summary on the test split.

Policy	Mean cost	Median cost	Solve rate	Timeout rate
VSIDS	406.7354	50.8760	0.9773	0.0227
MOMS	706.5590	29.7320	0.9318	0.0682
Jeroslow–Wang	684.6435	33.9140	0.9545	0.0455
score-with-R	797.8512	53.6500	0.9318	0.0682
frozen gate v1.7	778.2738	55.4080	0.9318	0.0682
Gate+L	791.5120	48.8220	0.9318	0.0682
Gate+shuffled-L	784.6958	50.6620	0.9318	0.0682

6 Figures

The generated figures are included for auditability and visual inspection. They are derived from the committed CSV outputs and contain no raw DIMACS clauses.

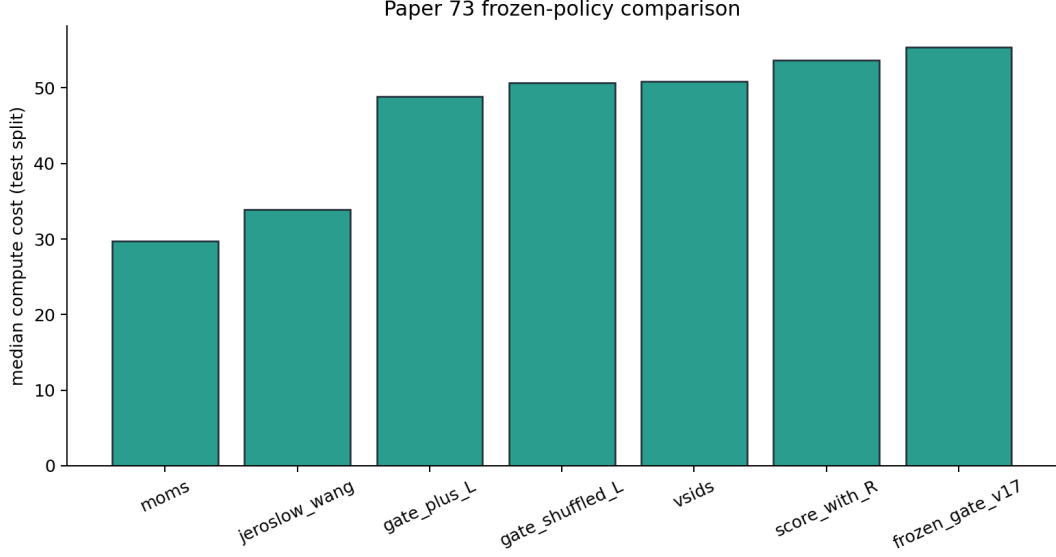


Figure 1: Median compute cost by policy on the test split.

7 Interpretation

The execution result is

`no_minimum_support`.

The preregistered Local Occurrence Hypothesis was tested under the predefined protocol and was not supported.

This negative result should be read narrowly. It does not show that local SAT structure is irrelevant, and it does not reinterpret the Paper-72 protocol. Rather, it shows that the specific frozen occurrence/degree residual channel defined in Paper 72 did not provide robust practical decision utility under the first external execution.

Central execution observation. The near-collinearity of the residualized occurrence and degree channels after calibration-only residualization rendered $L_\star$ effectively inactive in this execution.

The detailed numbers in the frozen feature output confirm this observation. The maximum absolute difference between O_{raw} and D_{raw} was 2.08×10^{-10} , and the maximum absolute difference between O_{res} and D_{res} was 1.14×10^{-10} . Consequently $L_\star$ was effectively zero (standard deviation 3.53×10^{-9}), and $G_{\text{gate+L}}$ remained practically indistinguishable from the frozen gate (maximum absolute deviation 6.19×10^{-10}). In this execution, the predeclared local channel therefore did not become an additional operative branching signal.

This sharpens the scientific reading of the negative result. The `no_minimum_support` classification is correct under the frozen criteria, but the execution should not be read as a decisive test of all possible occurrence/degree local-channel ideas. It tested the specific Paper-72 operational construction, and that construction collapsed onto a nearly redundant pair of channel measurements in this SATLIB implementation. Thus the frozen hypothesis was tested faithfully, but the added channel did not carry independent signal in the executed form.

Two outcome-level observations are especially important. First, Gate+L did not robustly beat the frozen v1.7 gate or score-with-R. Second, Gate+L did not survive the shuffled- L null. Together, these outcomes rule out minimum support under the predeclared criteria. The negative comparison against MOMS further indicates that classical decision-local short-clause pressure remains a stronger SAT branching signal in this benchmark.

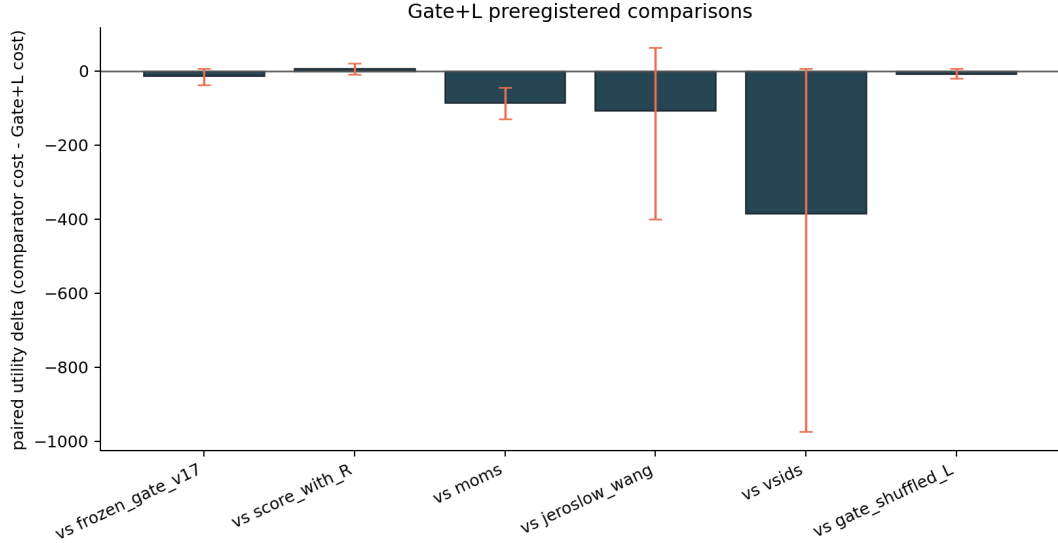


Figure 2: Paired Gate+L comparisons with bootstrap confidence intervals.

8 Limitations

This paper reports one frozen execution, not a new search for better SAT features. The benchmark is SATLIB-based and family-balanced only at the selected-family level. The transparent solver backend is useful for auditability but is not a substitute for highly optimized modern CDCL implementations.

No new hypothesis is introduced here. Any future local-channel refinement would require a new preregistered protocol and a new execution paper.

9 Reproducibility

The experiment folder is:

`experiments/local_occurrence_execution_paper73/`

The repository is:

<https://github.com/Chris4081/structural-selection-principle>.

The preregistered Paper-72 protocol artifact used by this execution is:

`../local_occurrence_hypothesis_paper72/
outputs_paper72_local_occurrence_protocol/paper72_protocol.json`

Raw SATLIB CNF files are intentionally not committed. They must be staged from the original SATLIB source and verified against the SHA256 manifest before any solver execution:

```
python3 download_satlib_paper73.py --allow-insecure-ssl
python3 local_occurrence_execution_paper73.py --validate-only
```

The insecure-TLS flag records provenance metadata only and does not affect gate parameters, residualization, calibration, policies, budgets, or interpretation criteria.

The frozen execution is reproduced by:

```
python3 local_occurrence_execution_paper73.py
```

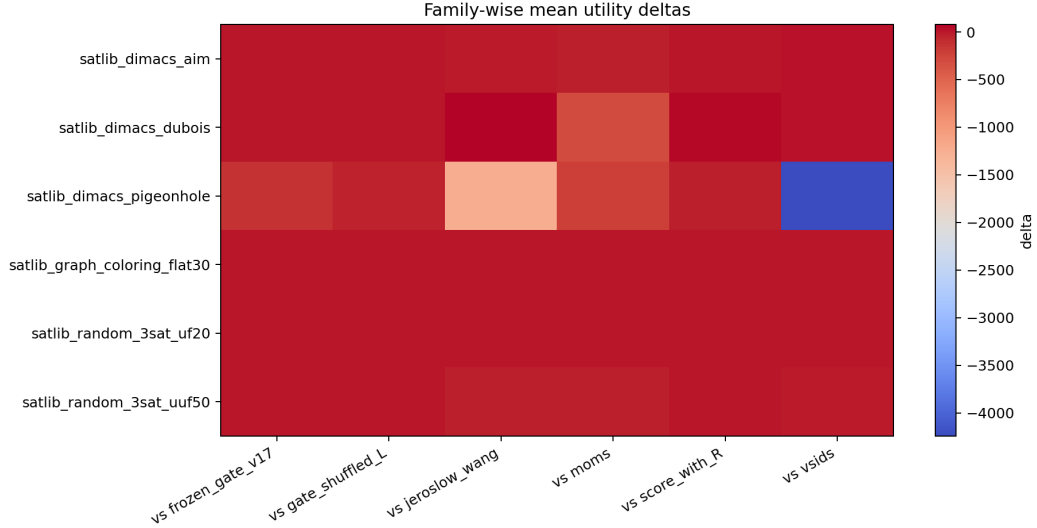


Figure 3: Family-level paired utility deltas. The strongest negative effects are concentrated in pigeon-hole and classical heuristic comparisons.

Primary outputs are written under:

`outputs_paper73_local_occurrence_execution/`

and include:

- `paper73_dataset_validation.json`,
- `paper73_summary.json`,
- `paper73_solve_records.csv`,
- `paper73_gate_l_features.csv`,
- `paper73_gate_l_comparisons.csv`,
- `paper73_policy_summary.csv`,
- `paper73_family_results.csv`,
- `paper73_shuffled_l_null.csv`.

The committed CSV/JSON files and figures are derived artifacts. They do not contain raw DIMACS clauses and do not redistribute SATLIB benchmark instances.

10 Conclusion

Paper 73 completes the first execution of the Local Occurrence Hypothesis defined in Paper 72. The dataset gatekeeper passed, the solver pipeline ran, and the output is reproducible from the committed code, manifest, and derived CSV/JSON files. Under the preregistered interpretation rule, the hypothesis receives `no_minimum_support`.

The scientific value of the result is methodological: the MAAT Gate Challenge cycle generated a concrete hypothesis, froze it, executed it externally, and accepted the negative outcome without modifying the preregistered hypothesis.

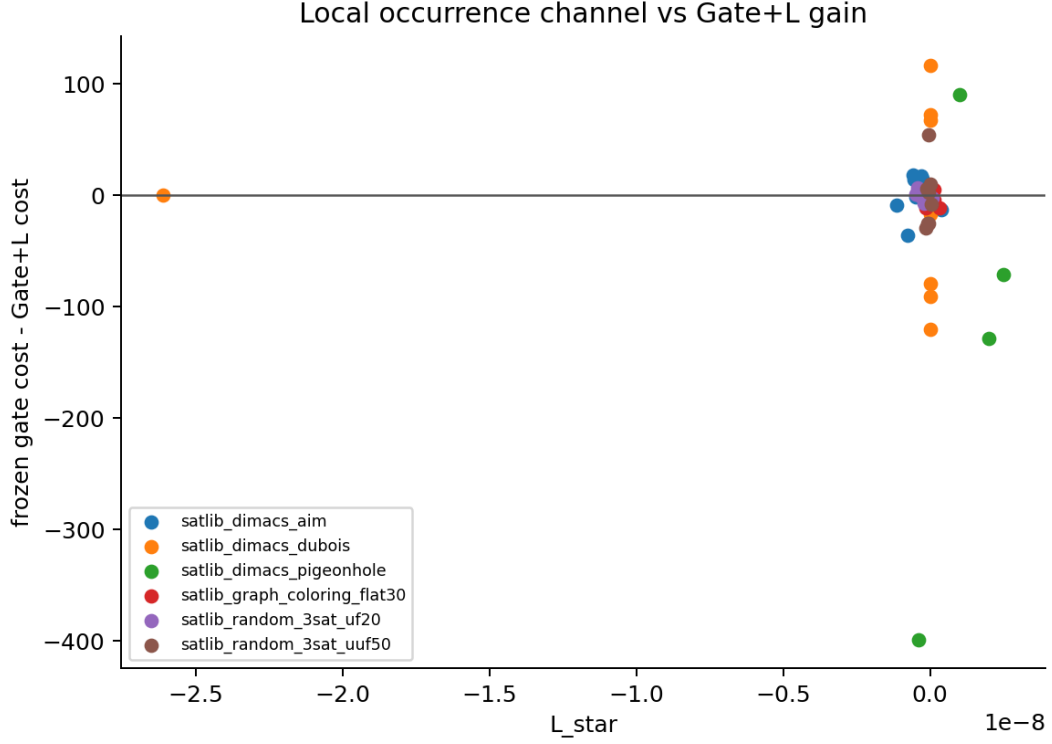


Figure 4: Gate gain against $L_{\star}$ on the test split. The local occurrence coordinate does not produce a robust positive execution signal under the frozen protocol.

References

- [1] H. H. Hoos and T. Stützle, *SATLIB — Benchmark Problems*. Public benchmark page, accessed 2026. <https://www.cs.ubc.ca/~hoos/SATLIB/benchm.html>.
- [2] M. W. Moskewicz, C. F. Madigan, Y. Zhao, L. Zhang, and S. Malik, “Chaff: Engineering an Efficient SAT Solver,” in *Proceedings of the 38th Design Automation Conference*, 2001.
- [3] R. G. Jeroslow and J. Wang, “Solving propositional satisfiability problems,” *Annals of Mathematics and Artificial Intelligence*, 1, 167–187, 1990.
- [4] C. Krieg, *MAAT Research Series II — Gate Challenge Cycle I*. Zenodo, 2026. <https://doi.org/10.5281/zenodo.21062386>.